



DPUSGString

DPUSGString presents a user-defined text string in the simulation. The text can even be multi-line.

Two different display modes are supported by the DPU:

- 2D mode (overlay): The text is inserted as an overlay at a fixed position and size of the view window. An example is a head-up display.
- 3D mode: The text is inserted at a 3D world position into the simulated environment and can be used to label an object. It can also be moved during the simulation.

1. Inputs:

<i>Format</i>	If <i>Format</i> =0 (the default), presents a simple single-line text. If <i>Format</i> =1, presents a single- or multi-line text into which a numerical input value is optionally inserted. Line breaks and the insertion are controlled by format strings (see section below).
<i>String</i>	Text to display. If <i>Format</i> =1 is set, special format strings can be used (see below).
<i>Number</i>	Number which is inserted into the text.
<i>PosX</i>	X coordinate of the position where the text is displayed (default = 100).
<i>PosY</i>	Y coordinate of the position where the text is displayed (default = 100).
<i>PosZ</i>	Z coordinate of the position where the text is displayed (default = 5). Please note that in the 2D overlay mode, only the range of 0 to 10 is visible. 10 is "in front" or "on top" of all other scene elements.
<i>ScaleX</i>	Output scaling in X direction (default = 1.0).
<i>ScaleY</i>	Output scaling in Y direction (default = 1.0).
<i>ScaleZ</i>	Output scaling in Z direction (default = 1.0).
<i>RotX</i>	Rotation about the X axis (roll angle) (default = 0.0).
<i>RotY</i>	Rotation about the Y axis (pitch angle) (default = 0.0).
<i>RotZ</i>	Rotation about the Z axis (yaw angle) (default = 0.0).
<i>ColR</i>	The RGB color of the displayed text.
<i>ColG</i>	For each channel, the range 0.0 ... 1.0 is valid.
<i>ColB</i>	Default: (R,G,B) = (0,0,0) i.e. black.
<i>ColA</i>	The opacity of the displayed text. The range is 0.0 (completely transparent) ... 1.0 (completely opaque). The default is 1.0



<i>Font</i>	Resource name of the font that is used for the text display. A font must be used that has been created using the ERSFontMaker application. A number of fonts are included with SILAB's scope of delivery in the folder (D:\SILAB\lib\graphics\default\texture\)\ERS.Fonts. Default: <i>ERS.Fonts.TimesNewRoman</i>
<i>FixedSize</i>	For the 3D display mode, selects if the text appears at a fixed size independent of its distance to the camera (<i>true</i> , the default setting) or if the text is scaled along with all other scenery objects (<i>false</i>).
<i>Billboard</i>	If enabled, for the 3D display mode, rotates the text object always perpendicular to the Ego driver's viewing direction, so that it is always readable. Default: <i>true</i> .
<i>Show</i>	Selects if the text is currently visible or not. (default: <i>true</i>).
<i>AL_HCENTER</i>	If set to <i>true</i> , aligns the center of the text area to the specified position. The default value <i>false</i> aligns the left border of the text area to the specified position.
<i>AL_RIGHT</i>	If set to <i>true</i> , aligns the right border of the text area to the specified position. The default value <i>false</i> aligns the left border of the text area to the specified position.
<i>AL_VCENTER</i>	If set to <i>true</i> , aligns the vertical center of the text area to the specified position. The default value <i>false</i> aligns the top of the text area to the specified position.
<i>AL_BOTTOM</i>	If set to <i>true</i> , aligns the bottom of the text area to the specified position. The default value <i>false</i> aligns the top of the text area to the specified position.

2. Static parameters:

<i>Overlay</i>	Select the display mode: Overlay = <i>true</i> : 2D mode (the default), Overlay = <i>false</i> : 3D mode.
<i>Visibility</i>	Visibility category for the object. It can be used to display the object on one of the views updated on a single graphics computer only.

3. Format strings:

If the Format parameter is set to 1, some special format strings can be used in the *String* value.

The symbol *%f* or *%g* is replaced by the current value of *Number*.

Optionally, the precision of the display can be defined: For example, the specification *%7.2f* displays the number using 7 characters, whereof 2 digits are after the decimal point.

Additionally, the symbol *%%* can be used to insert a percent sign, and *%n* can be used to insert a line break.

4. Dependencies to other DPUs:

DPUSGString must run together with **DPUSGEWorld**, and DPUSGString's Index value must be higher.



5. Example:

The following script is a minimal example with which DPUSGString can be used:

```
DPUSGString tach
{
    Computer = {LOCALHOST};
    Index = 350;
    Format = 1;
    String = "%0.0f km/h";
    Font = ERS.Fonts.Arial;
    AL_RIGHT = true;
    ColR = 1.0;
    ColG = 0.50;
    ColB = 0.250;
    ColA = 0.70;
    PosX = 960;
    PosY = 740;
};
Connections =
{
    VDyn.v_kmh -> tach.Number
};
```

This Configuration will display the current speed of the vehicle on the screen at *PosX* = 960 and *PosY* = 740. The speed value which we get from the Vehicle Dynamics *VDyn* is connected to the *Number* to be displayed. Since *Format* is set to 1, *%f* in the *String* to be displayed is replaced by the value of *Number*.